

The four w's: **What would Maria do?**

1 –What is the problem?

- Plastic on the beach and in the ocean
- Dangerous for animals and people
- Excess garbage with no use

Who experiences it?

- People who go to the beach and swim in the ocean
- Animals on the beach and in the ocean

Where does it occur?

- On the beaches and in the ocean in Rio de Janeiro

Why is it a problem?

- Bad for the environment
- Animals might mistake plastic for food
- Ruins natural beauty

2



I'm going to use the Four Ws method to create a problem statement

Problem Statement

I want to find a way to reuse the plastic on the beaches in Rio de Janeiro, so that we can decrease the amount of garbage on the beaches and make it safer and better for people and animals.

3

Draw or find a picture and insert it in the circle.



Think about their age, job,
hometown, health, income, etc.

Think about their hobbies,
interests, dreams, etc.

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Think about what they care about and the challenges they face

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Personas: What would Maria do?



I'm going to create 2 personas. They are part of the same target audience which is *people who go to the beach*.

Their key characteristics might be very different but their pain points around the issue can be similar.



Fatima

“Surfing is my meditation, a moment to be myself and one with nature”

Key characteristics:

- 21 years old
- Student
- Surfer

Interests/goals:

- Competitive surfing
- Environmental sustainability

Painpoints:

- Plastic in the ocean, sometimes confuses them for jelly fish
- Pollution



Nelson

“There is nothing quite like watching the sunset from a beautiful beach”

Key characteristics:

- 76 years old
- Retired
- Father of 4, dog owner

Interests/goals:

- Taking his dogs for long walks
- Picnics at the beach with his family

Painpoints:

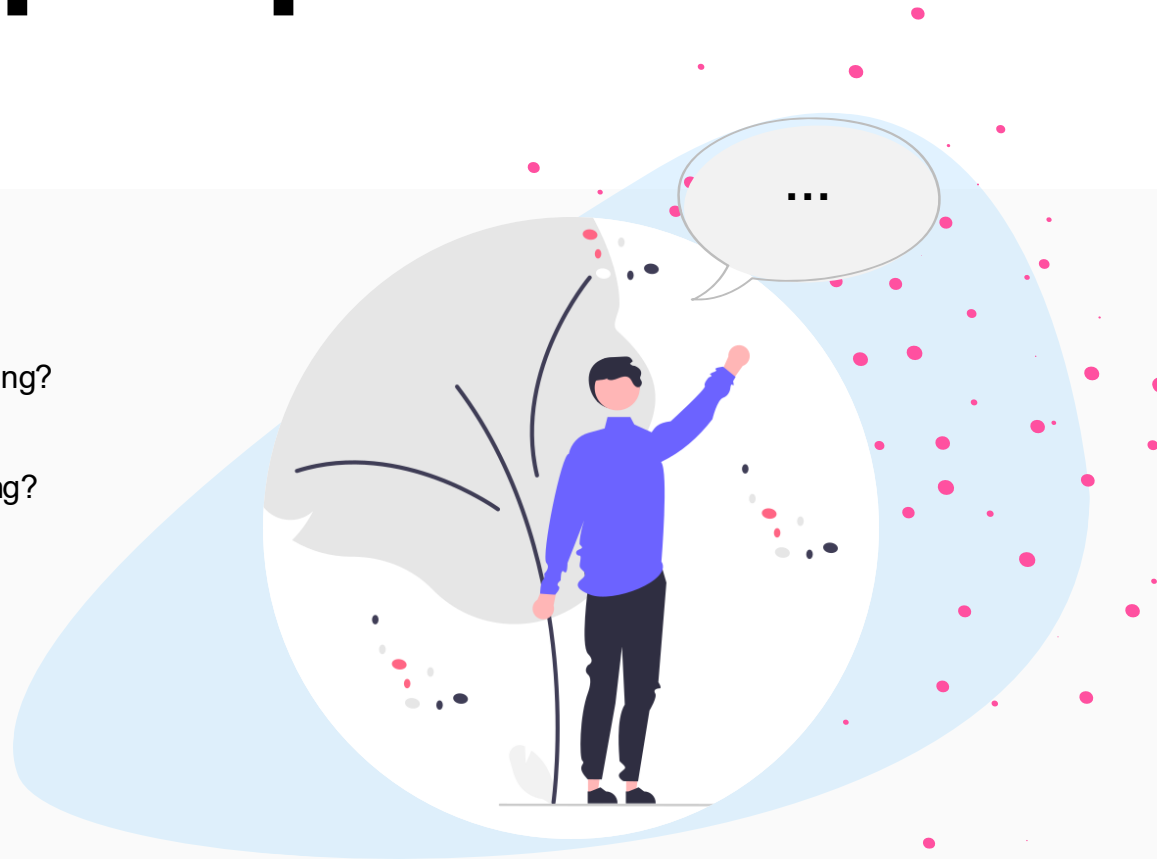
- His dogs always finding garbage on the beach
- Pollution

Empathy map template



Say & Do

- What do they say?
- What can you imagine them saying?
- What do they do today?
- What can you imagine them doing?



Think & Feel

- What do they think but keep to themselves?
- What are their needs, wants, hopes and dreams?

Hear & See

- What are they hearing others say?
- What are their friends saying?
- What are they hearing from their colleagues?
- What do they observe others doing around them?
- What do they see in their immediate environments?
- What are they watching and reading?

Positives

- What is improving their experiences and life?
- What works well for them?
- Where or when are they happiest?

Pains

- What frustrations do they have?
- What about their experiences is disappointing?
- What are their fears?

Empathy map: What would Maria do?



Say & Do

- They enjoy spending time outdoors in nature
- They are active
- They have to decide if they want to find other ways to have fun and relax
- They enjoy spending time with their family.

Hear & See

- The beach used to be a place where I could relax but the plastic pollution causes me stress
- We need to recycle more
- Throwing rubbish in the wrong places
- A need for recycling and waste prevention
- They are watching nature shows

Positives

- Being able to relax and spend time with their family
- Spending time with friends
- At the beach when it is clean and serene

I'm going to put myself in my users shoes to think about the problem from their perspective.

Think & Feel

- There should be penalties for individuals that litter
- They feel sad
- For animals to be safe and for them to have a good experience going to the beach

Pains

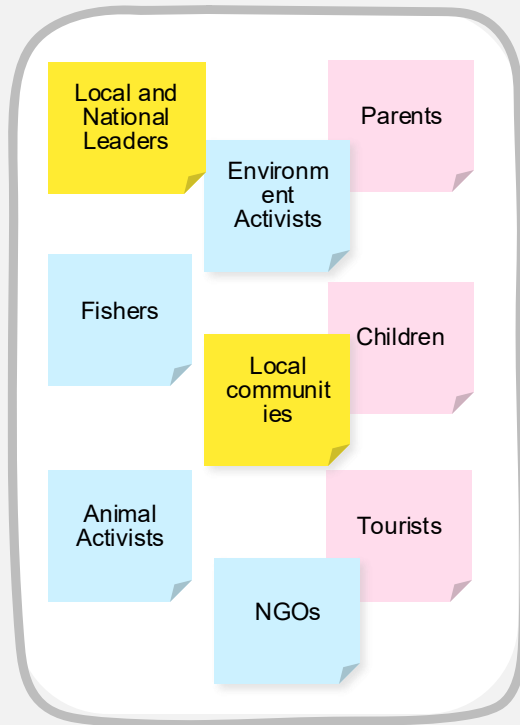
- Pollution and waste
- Plastic pollution not only ruins beach experiences but is also dangerous to animals.
- The plastic pollution getting completely out of control



Stakeholder mapping: What would Maria do?

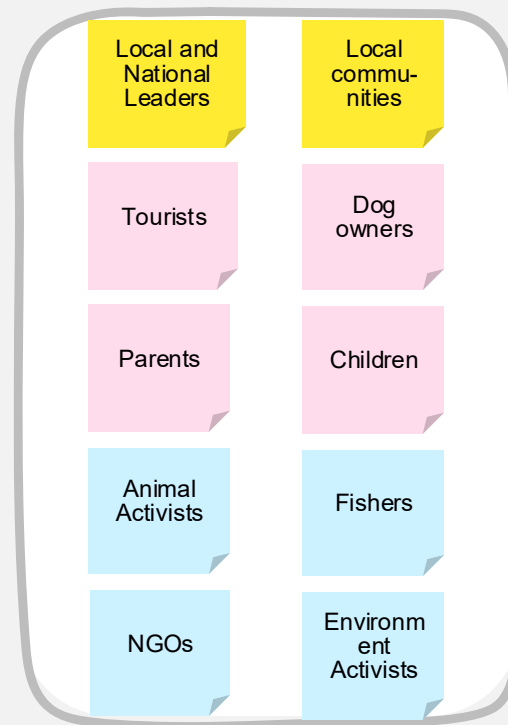
Brainstorm who your stakeholders are

Write down each different stakeholder on it's own piece of paper.



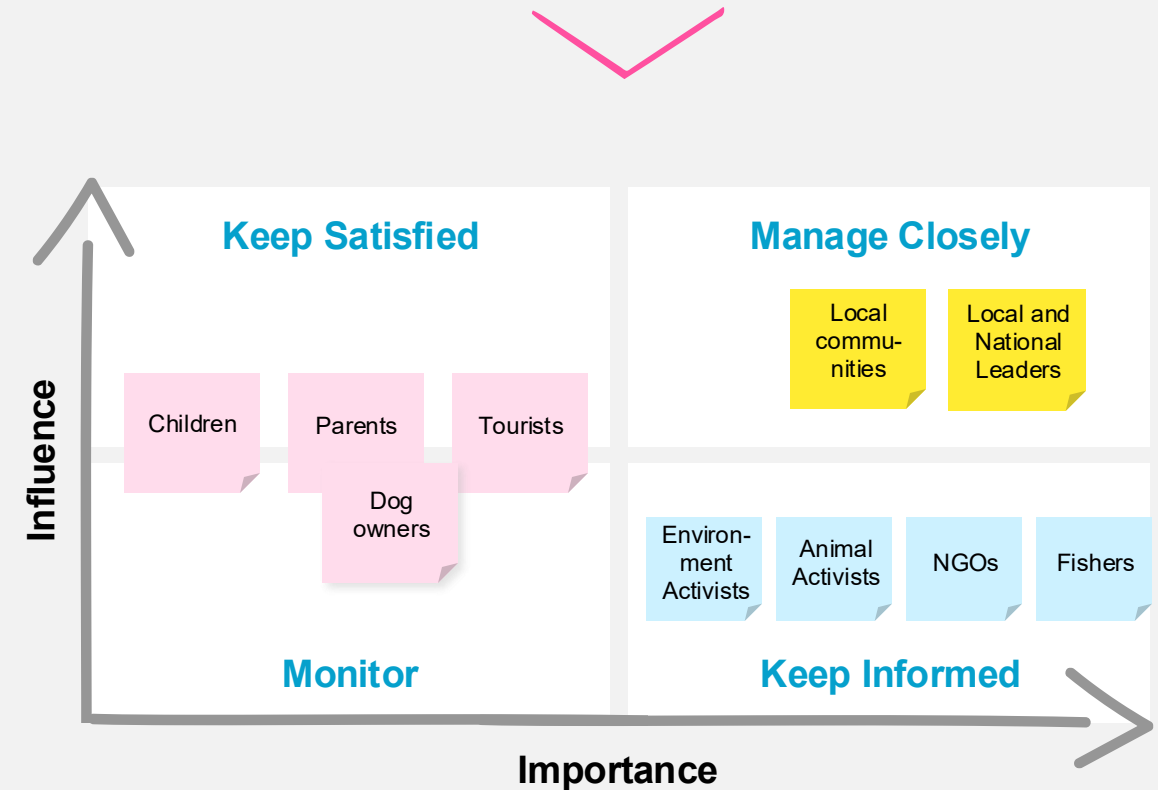
Group similar stakeholder together

Put the pieces of paper with stakeholders who are similar in groups



Prioritize your list of stakeholders

Draw an outline of the chart below and decide where each of the stakeholder groups should go based on their influence and importance to the project



Crazy 8s: What would Maria do?



Idea
1

Organize community
beach cleans



Idea
2

Create surfboards
from recycled plastic



Idea
3

Create children's toys
from recycled plastic



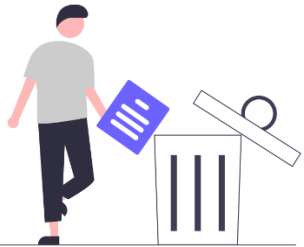
Idea
4

Create umbrellas
from recycled plastic



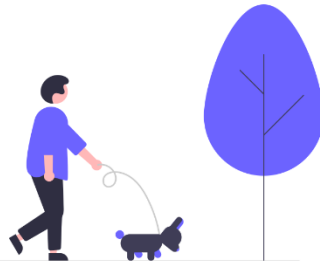
Idea
5

Get more garbage
bins on the beaches



Idea
6

Create dog toys
from recycled plastic



Idea
7

Create fishing rods
from recycled plastic



Idea
8

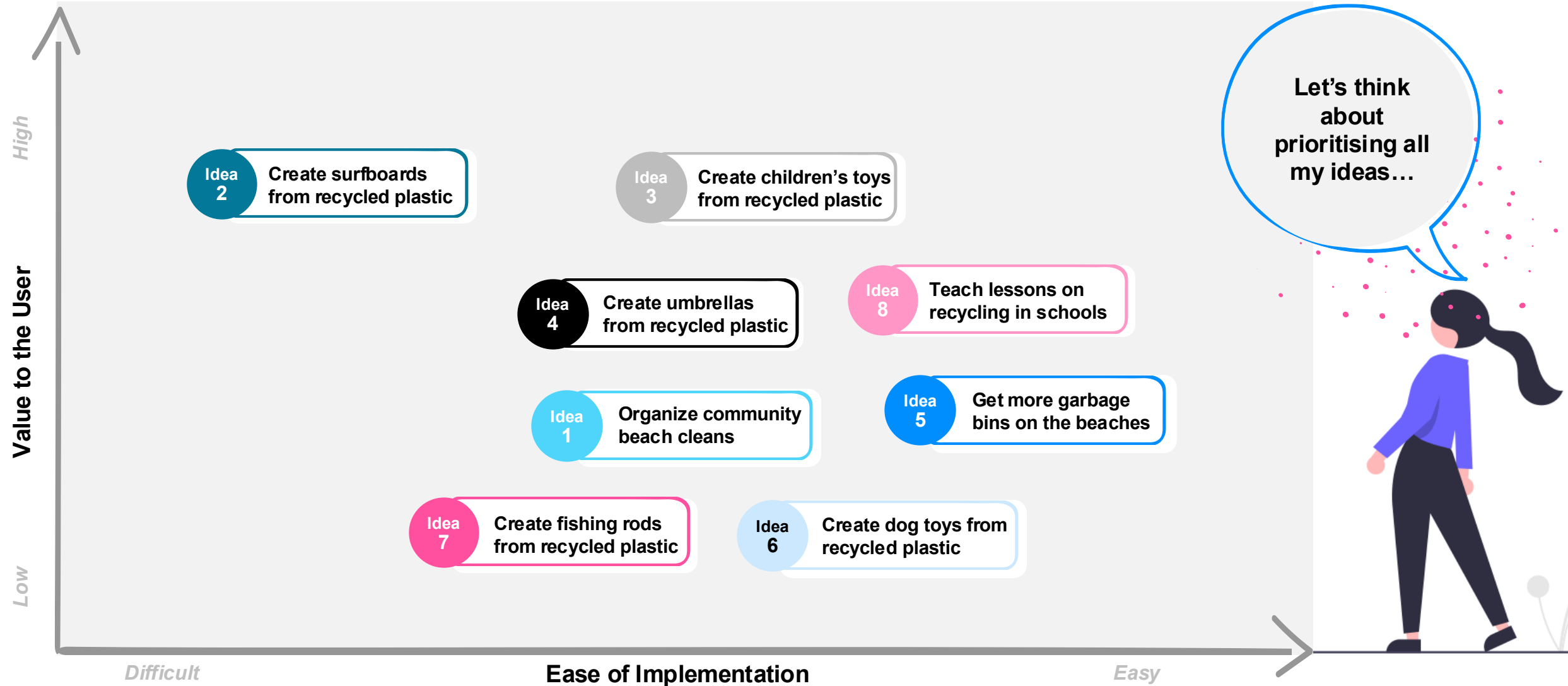
Teach lessons on
recycling in schools



Let's see
what ideas
I can think
of in **8 min**
...



Prioritization grid: What would Maria do?



NABC: What would Maria do?



Let's evaluate my idea using the Needs, Approach, Benefits and Competition tool

Need

Plastic Pollution and the potential that they will grow up in a damaged and polluted environment

A need to educate the younger generation so they don't make the same mistakes of the past

The future generation

This problem happens everywhere but we are focusing on beaches and the ocean



Approach

Developing a children's toy from recycled plastic waste



Create children's toys from recycled plastic

This product will aim to educate, empower and enable children to understand the importance of recycling and looking after the planet.

Plastic sanitisation, toy designers and toymakers



This will ensure there is a purpose for the existing plastic



Fun, education and awareness

Helps parents guide their children

This aligns with the YIF goal of addressing the barriers that young people face to productive, quality livelihoods in their local communities

Benefits



Recife has a community of women that sell handbags, jewellery and toys out of plastic waste

Going into schools to educate young children

This idea also aims to educate

Competition

Idea clusters: What would Maria do?



First, let's write down all my ideas...

Now let's group them based on similar themes!

Idea 1 Organize community beach cleans

Idea 5 Get more garbage bins on the beaches

Idea 2 Create surfboards from recycled plastic

Idea 6 Create dog toys from recycled plastic

Idea 3 Create children's toys from recycled plastic

Idea 7 Create fishing rods from recycled plastic

Idea 4 Create umbrellas from recycled plastic

Idea 8 Teach lessons on recycling in schools

Theme 1: Beach Activities

Idea 1 Organize community beach cleans

Idea 5 Get more garbage bins on the beaches

Theme 2: Education

Idea 8 Teach lessons on recycling in schools

Theme 3: Toys

Idea 3 Create children's toys from recycled plastic

Idea 6 Create dog toys from recycled plastic

Theme 4: Beach products

Idea 2 Create surfboards from recycled plastic

Idea 4 Create umbrellas from recycled plastic

Idea 7 Create fishing rods from recycled plastic



Lean canvas: What would Maria do?



1

What problems have you identified?

- Plastic pollution
- Unsafe environment for people and animals
- Lack of education on sustainability
- Global warming

Existing Solutions

- Recycling
- Beach cleans
- Sustainability education initiatives

2

What solution have you come up with?

Children's toys:

- Made from recycled plastic found on the beaches
- Each toy will be based on an ocean animal and will include a story about the animal and why it's important to protect it
- Kids are encouraged to collect all the toys so they can create their own ocean habitat and learn about each of the beautiful ocean animals

3

Describe 3 different types of people who might use your solution

- Children aged 5-10
- Love animals and the ocean
- Product bought by parents/relatives

How do you reach these people?

- Online advertisements
- Sustainability blogs for parents
- Online and physical store
- Sell on the beachfront

4

What makes your solution stand out?

It's both a fun toy and a way for kids to learn about different ocean animals and why they are important for our environment

Can your idea be easily copied by others?

Not difficult to copy. Need strong branding and important to tell the story of how we use recycled plastic to create the toys.

5

How would you know your solution works? How do you measure your success?

- Number of toys sold
- Increase in sustainability awareness in kids (based on their feedback & engagement)
- Number of pieces of plastic recycled

6

*How can you measure impact and sustainability?

- Measure the amount of plastic we collect from beaches and recycle into toys
- Number of kids who buy the toys and learn from them
- Get feedback from kids on what they've learnt from each toy, see if we can improve this with new toys

7

*How much does it cost to make your solution work?

- Cost of time and labor to collect plastic
- Cost for toy design and development
- Advertising costs

I want to get an overview of my idea to create kids toys from recycled plastic!



Roadmap: What would Maria do?



The timeline we want to work with:

Team members:	Month 1		Month 2		Month 3		Month 4	
Name: Maria	Find toy manufacturers		Select final toy manufacturer		Oversee management of toy manufacturing		Do quality checks	Launch first toys!
Name: Peter	Create strategy to collect plastic		Collect plastic from beaches to be reused		Provide plastic to manufacturer	Continue to source plastic	Measure amount of plastic reused	
Name: Louisa	Speak to kids on toys they like	Start initial toy designs	Input kids feedback into toy designs	Finalise 2 toy designs	Work with manufacturer and Maria to get toys made		Review feedback and iterate	Design next range of toys
Name: Carlos	Start creating branding, website, social media		Contact potential sellers of toys	Create online store on website	Share content of beach cleans	Plan strategy for adverts	Get toys into stores	

There's three other people on my team. We want to launch our first product in four months!

Let's plan how me and my team will work together to get our children's toys created!



User journey mapping: What would Maria do?



Adriana

Key characteristics:

- 10 years old
- Enjoys playing
- Loves the beach

Interests/goals:

- Wants to be a marine biologist
- Sustainability

Painpoints:

- Plastic on the beaches
- Sad when she thinks about animals eating plastic
- Not enough toys

I'm going to map the user journey of a persona called Adriana



Action

Define what actions your user takes when they use your product/service. This could include what they do before and after they interact with it too! These actions are the key stages in your user's journey.

Thoughts

Write what you imagine your user is thinking at each stage of their journey.

Feelings

Use faces like the below to show how your user is feeling over each stage.



Opportunities

Think about what opportunities there are to improve your user's experience at each stage.

She goes to the beach and finds plastic everywhere. Sees a bird eating it.

This is sad. Animals shouldn't eat plastic.



Recycling bins and information available at the beach.

She tells her parents she wants to learn more about sustainability.

I'm interested in sustainability but don't know how to learn about it.



Learning materials available on our website for free.

She receives one of our toys as a gift from her parents for her birthday.

Excited to play with my toy and learn at the same time!



The toy could speak and tell facts about sustainability.

She plays with her new toy, reading about how it is made & learning about the animal it is.

It's so cool that it's made from recycled plastic. I've learnt a lot.



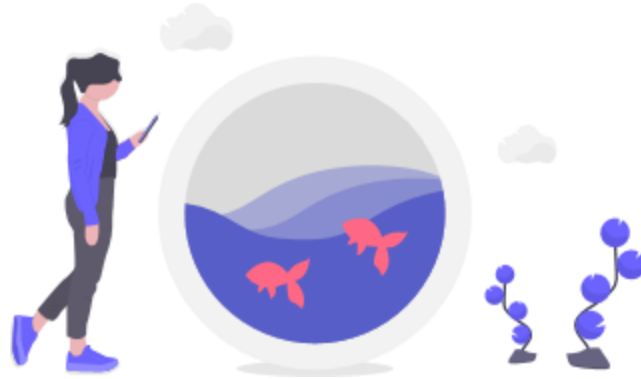
Share more about how we collect plastic and create toys.

Paper prototype: What would Maria do?



Let's draw a prototype of my toy on a piece of paper!

Draw a sketch of your solution in the box below



Research animals found in the ocean and near the beaches who would be impacted by plastic pollution.



Design toys based on these animals. Use the plastic collected on the beaches to create the toys.



Sell the toys to kids with educational information on how they are made, and why it's important to protect these animals and our environment.

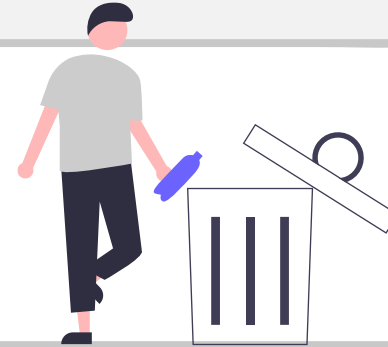
Storyboarding prototype: What would Maria do?



I'm going to draw my solution and the steps it takes to create my end product.



Go to beaches



Collect plastic waste



Deliver plastic to toy manufacturer to be reused for toy creation



Sell toys to children with information on each animal and their importance to our environment



Kids learn about sustainability by playing with the toys and engaging with learning materials



Live prototyping: What would Maria do?



Use this template to plan your Live Prototype.

Let's plan how the Live Prototype of my toy will go!



What do you want to test? What are the main objectives of the Live Prototype?

How children feel about the toys, if they have fun playing with them and if they learn about sustainability while using them.

List the Key Stakeholders that will be involved?

- Children
- Parents/relatives of children

How many users will take part?

- 10-15 children plus their parents/guardians

What do you need to make this possible? Think about: A physical space, volunteers, resources

- A safe and welcoming space for children to play
- A prototype of the toy for children to play with and provide feedback on
- Team members who can ask children to share their experiences

How long will the live prototype take?

Can it be done in 1 day or less? Will it take longer and take place at different times over a few weeks?

- 1-2 hours of play and feedback time

How will you record and capture feedback?

- Observing children playing with toys to see if they like them
- Asking them questions about what they learnt about each animal and sustainability after the play time is over

How will you ensure feedback is considered in the development of your solution?

- We will identify what things kids did learn about sustainability and see how we can expand on this with the product
- Ask kids what animals they'd like to learn about in future products

‘I like, I wish, I wonder’: What would Maria do?



Let's find out ways to make the product better!

Thing you want feedback on

What children and parents think about the toys.

If children enjoy playing with them and if they learn about sustainability while using them.

Can any improvements be made on the toys.

I Like...



Toys are made from recycled plastic

The stories around each toy

It creates a wider story of the ocean

Makes topical issues accessible to children

I Wish...



There was a storybook to go with the toys.

It had info about local charities

I Wonder...



If parents and children could join beach clean ups

Is this in partnership with any local charities?

Can there be visits to local schools to share about this

Reflecting on the Feedback

Parents and children want to apply the knowledge they have learnt and make an impact through volunteering opportunities in their local areas. This is something that can be featured on the website to include a “Beach Clean near you” section. There is also an interest in partnering with schools and local environmental charities. School partnerships will definitely be explored in the long term and partnerships with local environmental challenges will be highlighted on all digital and physical information.

The mini test: What would Maria do?



Test Preparation

Brief description of the Test Situation: Think about where the Test is happening, which user(s) are involved, how long will it last?

The test will take place at an after-school club with children, teachers and parents that can attend.
The test will last for 45 minutes.

Test Roles: Who is going to be involved in running the test? What are their responsibilities?

The product team and teachers. To clearly explain the test to children and parents and facilitate the running of the test.

How will you record and capture feedback?

With survey sheets that will be completed by teachers and parents. And in group discussions that will be noted.

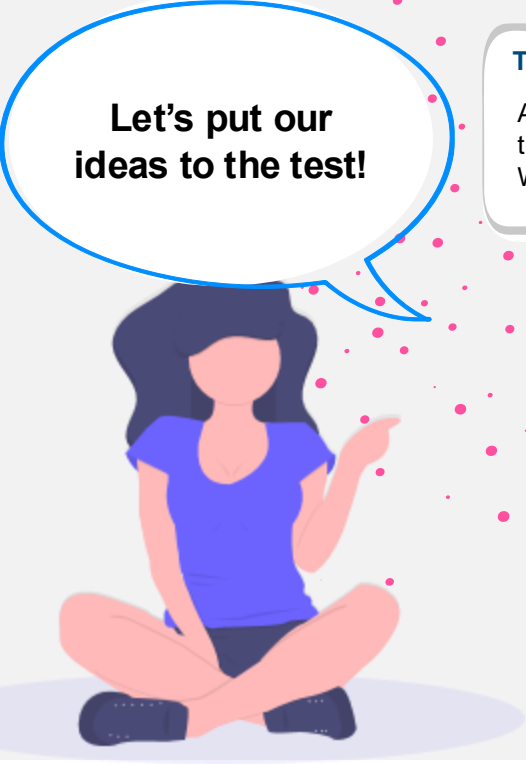
Test Results

Test Documentation: Summarize the main findings and learnings from your Test

All the children had positive feedback from the test session, the main thing that they liked was the educational aspect. Through this session alone, we were able to see an increased confidence in what they children understood about the ocean and the environment. One feedback that we received is that the story element of the toys looked like it was for parents and not for children. Also, a parent suggested having different versions for different ages, the toy can remain the same, but the materials offered with he toy should be more tailored to specific ages.

How will you ensure feedback is considered in the development of your solution?

By sharing the feedback with the team and finding ways to implement it into the product design.



Let's put our ideas to the test!

Solution interview: What would Maria do?



1

Interview Goal

Does this product educate children on the dangers of plastic pollution and increase sustainability awareness, in a fun and interactive way?

3

Plan the Interview

2 Interviews per user - The first interview will be before using the toy and the second interview will be after using the toy.

1st Interview – What do you know about the ocean? Where have you learnt this? How would you learn more?

2nd Interview - What do you know about the ocean? What did the toy teach you? How did you find the toy?

Create the Interview Agenda

1. Drinks and Snacks
2. Icebreaker
3. Intros
4. Play video about the project
5. Ask questions
6. Summary and Close

It's important to understand if the product that we are building fulfils its purpose.

2

Pick the People to Interview

6 parents and children that cover the age range of the target user.

4

Run the Interview

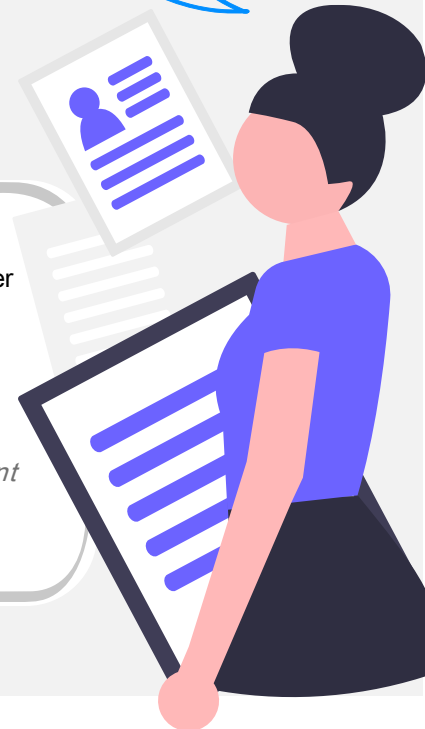
Offering drinks and snacks, making sure it's not across the table.
Decorating the room in a child friendly way.

5

Review Insights and Refine Your Solution

All parents and children offered very positive feedback on the toys. After the first interview when the children had an opportunity to play with the toys, there was a significant improvement in their knowledge and confidence in facts about the ocean, sustainability and plastic pollution.

Parents suggested having a parent/teacher handbook alongside the children's information sheets to make it more interactive. Another parent suggested including some colouring pages for children that enjoy arts and crafts as part of the storytelling elements.



Create a pitch: What would Maria do?



Beginning with a Story

Meet Nelson...
Nelson is a dog owner and a Father of 4.

He lives close to the coast and enjoys taking his dog out for walks at the beach and having beach days with his family.

Recently on their walks, his dog keeps finding plastic waste at the beach. Also, Nelson has found that his local beach is getting messier and finding a spot for his family picnic is not as easy as it used to be.

Problem

8 million tonnes of plastic waste ends up in the ocean every year.

That is more than the weight of 2 million elephants combined.

This contamination is a serious problem as it takes over 400 years for plastic to decompose, so we must act now!

Users

This critical challenge doesn't only impact Nelson and other people who go to the beach but it also negatively impacts animals in the sea, the fishing industry and future generations.

How can I communicate my product in a clear way.

Solution

Children's toys made from recycled plastic found on beaches. Each toy will be based on an ocean animal and will include a story about the animal and why it's important to protect it.

Kids are encouraged to collect all the toys so they can create their own ocean habitat and learn about each of the beautiful ocean animals.

This solution is unique as it aims to educate children and offer age specific materials. It also offers a guidebook for parents and teachers on ways that they can use the toys and resources to educate children.

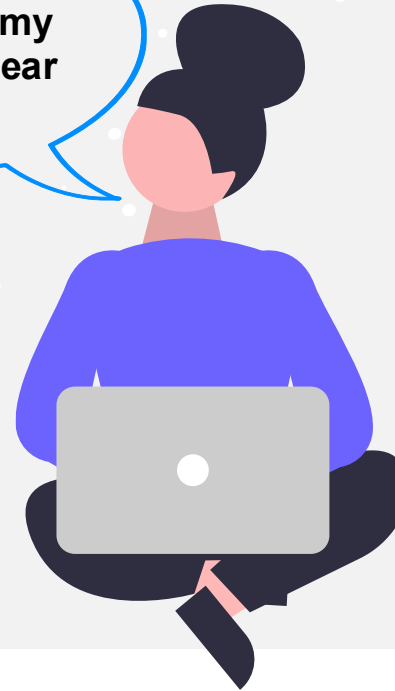
Business Model

This can be turned into a business by making a website for the toys to be purchased. Also partnering with local retailers that are close to the beach.

Challenges may be how long it takes for the toys to be developed but this can be overcome by finding a good toy manufacturer.

Next Steps

The first prototype has been built and we are now in a testing phase, in this phase we are getting feedback from potential users and we are implementing the feedback into the product..



Self reflection: What would Maria do?



What am I reflecting on?

A pitch the team delivered

When did this take place?

The pitch took place on 17 June and it lasted 10 minutes with 5 minutes of Q&A.

What are some of the key things I have learnt in this experience?

What went well?

All of the panellists really liked the concept and how we have implemented feedback in the process of developing the product

What could have gone better?

Answering the questions

What did I find most difficult?

Being able to explain the idea in 10 minutes and ensure no important information was missed out.

What am I most proud of?

Our ability to clearly explain the product and the goals of the product.

What would I do differently next times?

All of the panellists really liked the concept and how we have implemented feedback in the process of developing the product.

What are my future targets and goals?

Continue to build confidence in pitching the product, and develop future pitches based on the feedback gained from this pitch.

